Bihar Engineering University, Patna
VIIIth Semester Examination - 2023

Course: B.Tech. — *Time: 03 Hours*
Code: 105819 — *Subject: Adhoc and Sensor Networks* — *Full Marks: 70*

Instructions:-
- *(i) The marks are indicated in the right-hand margin.*
- *(ii) There are **NINE** questions in this paper.*
- *(iii) Attempt **FIVE** questions in all.*
- *(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) What is wireless communication?
(i) Sending data from one location to another with the use of physical medium
(ii) Sending data from one location to another without the use of physical medium
(iii) Sending data from one location to another without the use of virtual medium
(iv) None of the above

(b) Which of the following is not a fast fading propagation mechanism?
(i) Reflection (ii) Diffraction (iii) Refraction (iv) Scattering

(c) Wormhole attack
(i) receives packets at one location in the network
(ii) creates routing loops
(iii) selectively dropping packets
(iv) may leak confidential information to unauthorized users

(d) What is a defining characteristic of an ad-hoc network?
(i) Centralized control and management
(ii) Fixed infrastructure with wired connections
(iii) Dynamic and self-configuring nodes
(iv) Large coverage area

(e) What type of topology is commonly used in sensor networks?
(i) Star topology (ii) Ring topology
(iii) Mesh topology (iv) Bus topology

(f) Which networks are characterized by their dynamic topology and decentralized nature?
(i) Wired networks (ii) WSN
(iii) VPN (iv) Cellular networks

(g) In reactive routing (on-demand) protocol, a route to a destination is established:
(i) Periodically, at regular intervals
(ii) When a node detects link failure
(iii) In advance, before data transmission
(iv) Upon request, only when needed

(h) Which routing protocol is considered proactive and maintains up-to-date routing information at all times?
(i) AODV (ii) DSR
(iii) OLSR (iv) TORA

(i) In ad-hoc network routing, which type of routing protocols maintain routes to all possible destinations?
(i) Proactive (table-driven) routing protocols
(ii) Reactive (on-demand) routing protocols
(iii) Hybrid routing protocols
(iv) Adaptive routing protocols

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(j) In wireless ad-hoc network
(i) access point is must
(ii) access point is not required
(iii) nodes are not required
(iv) all nodes are access points

Q.2 What is a wireless sensor network? Elaborate the wireless sensor network architecture with a diagram. [14]

Q.3 (a) What are MAC protocol used in wireless sensor networks? [7]

(b) What are differences between wireless Adhoc network and wireless sensor network? [7]

Q.4 (a) Discuss the security attacks and security vulnerabilities in wireless networks. [7]

(b) Describe the key characteristics of MANETs and discuss their applications in real-world scenarios. [7]

Q.5 (a) Explain the key features and goals of the IEEE 802.15.4 standard. How does this standard contribute to the development of Wireless Sensor Networks? [7]

(b) How hybrid TDMA/FDMA medium access control protocol for wireless sensor network work? Explain with an example. [7]

Q.6 (a) Discuss the pros and cons of increasing the mobility margin in the power control scheme for hybrid wireless networks. In the power control scheme for hybrid wireless networks, what are the two important factors that should be considered to decide the value of the mobility margin? [7]

(b) Calculate the duty cycle and average power to peak power ratio of a UWB system in which the monocycle duration is 0.5 ns, pulse repetition interval is 25 ns and pulse amplitude is 0.1 mW. [7]

Q.7 (a) Wireless optical WDM rings provide high data rate networks in metropolitan areas. Discuss the possible solutions and factors to be considered for providing reliability for a wireless optical WDM ring network. [7]

(b) Discuss the effects of increasing the control zone radius in Meghadoot architecture in terms of the resource requirements at the IN and the routing efficiency. [7]

Q.8 Present a wireless sensor network design that can be used for surveillance and environment monitoring in a zoo. A zoo is a facility in which animals are confined within enclosures, displayed to the public, and in which they may also be bred. State the functional requirements you are considering. [14]

Q.9 Write short notes on ***any two*** of the following: [7x2=14]
(a) SPINS
(b) Preferred Destination-Ring Routing Scheme (PDRS)
(c) Software-based anti-temper techniques
(d) TCP over ad-hoc wireless networks

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